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Compost maturity prediction and gas emissions monitoring: A sensor-based and interpretable machine learning approach

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ABSTRACT

Here, a sensor-based machine learning approach has been presented to predict the maturity and monitor gas emissions during the composting process. By analyzing key environmental factors and emission data, our study aims to enhance the ecological responsibility of composting as a waste management solution. Our research combines a dedicated sensor system with machine learning. The sensor system, integrated with Arduino Mega 2560 R3 and ESP-32 microcontrollers, wirelessly transmits data for remote monitoring. Meanwhile, our machine learning framework analyzes features such as temperature, C/N ratio, ammonia concentration, pH levels, and nitrate content from ten datasets. After rigorous preprocessing and model training with a robust five-fold cross-validation, we optimize hyperparameters using GridSearchCV. The results highlight that both XGBOOST and CatBOOST excelled in achieving the highest predictive accuracy among the models, each attaining an impressive R^2 of 0.9912. In particular, XGBOOST demonstrated the lowest mean absolute error (MAE) at 1.1845, while CatBOOST exhibited the lowest mean squared error (MSE) at 1.8382. The interpretability of the model is ensured through LIME and SHAP, making complex models transparent and understandable. The results indicate that the XGBOOST model outperforms the others, achieving the highest predictive accuracy. This groundbreaking approach bridges scientific rigor with practical usability, ensuring responsible waste management for a sustainable future. Real-world applications of our research include more efficient and environmentally friendly waste management systems, reduced environmental impact, and improved compost quality for agricultural use.

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1. Introduction

There are various waste management methods, including incineration, landfills, and aerobic composting and anaerobic composting. In comparison to landfill and incineration, aerobic and anaerobic composting present more environmentally friendly approaches, especially for organic waste. Landfills and incineration release substantial amounts of greenhouse gases and pollutants such as carbon dioxide and methane, making them less eco-friendly options [1,2]. Aerobic composting takes place in the presence of oxygen and relies on microorganisms, such as bacteria, to decompose organic waste. In contrast, anaerobic composting occurs without oxygen and involves anaerobic microorganisms such as methanogens and lactobacilli for decomposition. Vermicomposting is an example of aerobic composting, which uses worms to break down organic matter, while bokashi composting exemplifies anaerobic composting, using a blend of effective microorganisms, such as lactic acid bacteria and actinomycetes, to break down complex organic compounds. However, both aerobic and anaerobic composting techniques release a variety of pollutants. Depending on the organic waste being processed, such pollutants can include methane, carbon dioxide, nitrogen dioxide, and sulfur dioxide [3]. Despite the emission of pollutants, both aerobic and anaerobic composting produce nutrient-rich compost that improves soil health, thus reducing the need for chemical fertilizers. These composting methods contribute to a more sustainable and environmentally conscious waste management approach [4].

1.1. Importance of monitoring gas emissions

It is crucial to monitor the release of toxic air pollutants and greenhouse gases, such as carbon dioxide (CO_2), methane (CH_4), nitrogen dioxide (NO_2), and sulfur dioxide (SO2), as they significantly impact the environment and human health. Both aerobic and anaerobic composting processes contribute to the emission of these harmful gases. Carbon dioxide is one of the primary contributing factors to global warming because of its ability to trap heat in the atmosphere of the Earth. NASA reported that in the future there will be a global increase in temperature within the range of 2 °C to 6 °C [5]. Consequently, this temperature increase will accelerate the melting of glaciers and ice sheets, as the reduction of ice cover leads to increased absorption of radiation from sunlight. As a result, there will be a rise in sea levels, as asserted by Latake et al. [6], with a projected increase in Earth's average sea level ranging from 0.09 to 0.88 m between 1990 and 2100. Additionally, methane, as the second most produced greenhouse gas, contributes significantly to climate change [7]. Moreover, it indirectly contributes to the production of ozone-depleting substances, ultimately leading to the depletion of the ozone layer [8]. These far-reaching consequences require careful monitoring and mitigation efforts.

Furthermore, the emission of toxic air pollutants, such as nitrogen dioxide (NO_2) and sulfur dioxide (SO_2), are equally urgent concerns. Emitted during composting, the pollutant degrades air quality, leading to respiratory problems such as bronchitis and asthma, as highlighted by D'amato et al. [9]. Additionally, high concentrations of NO_2 and SO_2 cause acid rain as these gases react with water vapor to form sulfuric and nitric acids [10], damaging soil and aquatic ecosystems.

By understanding and monitoring these emissions during composting, we can work toward mitigating their adverse effects on the environment and human health, ensuring that composting remains an ecologically responsible waste management solution. In doing so, we not only protect the environment but also preserve the well-being and prosperity of future generations.

1.2. IoT-based gas monitoring systems

To efficiently monitor the release of toxic pollutants, it is important to optimize data collection and exchange [11]. Internet of Things (IoT) services have improved the efficiency of connecting devices to sensors and monitoring systems over the Internet. To implement IoT-based gas monitoring services, it is crucial to have accurate monitoring and sensing systems [11,12]. Moreover, it is necessary to have a strong platform on which individual data from each gas sensor can be monitored, exchanged, and examined.

Thingspeak is a cloud-based platform that can be used for data exchange and management in such systems. Thingspeak can be used to store the data from gas sensors and present them in graphical forms. The data can then be stored, retrieved, and examined for trends and anomalies, which allows a deeper understanding of these toxic gases and the best ways to limit their production [12]. IoT devices are used to communicate sensor data to Thingspeak using the wireless Internet [13]. To maximize efficiency, devices such as the Raspberry Pi or Arduino are used for the data collection and preliminary processing of the data from each sensor. Consequently, a secondary Wi-Fi-based device, such as an ESP32, can transmit the data to the Thingspeak cloud where it can be graphed and monitored. This approach can greatly enhance data acquisition and processing for IoT-based systems [13]. After employing the data collection approach, we proceeded to train our model to predict composting maturity and enhance model transparency using tools like SHAP.

This research introduces IoT-based systems, highlighting efficient data collection and exchange, utilizing devices such as Raspberry Pi, Arduino, and ESP32 for enhanced data acquisition in sustainable waste management. The investigations are followed by a novel machine-learning approach for compost maturity prediction that is validated with external data in real implementation. Finally, a model transparency study has been conducted to enhance the model transparency, to employ interpretable machine learning techniques such as LIME and SHAP to provide insights into the decision-making process and to increase confidence in real-world applications. To the best of the authors knowledge, this study is one the limited research works that incorporates machine learning and IoT-based GHG monitoring.

This study focuses on addressing key environmental issues in composting by utilizing IoT-based monitoring systems and machine learning models. By integrating advanced technologies, the research aims to optimize the composting process while ensuring transparency and practical applicability. The main goals of our study is:

- Develop a sensor-based IoT system for real-time monitoring of gas emissions during the composting process.
- Utilize machine learning models to predict compost maturity with high accuracy.
- Enhance environmental sustainability by minimizing harmful emissions and optimizing composting practices.
- Provide interpretable insights using tools like LIME and SHAP to ensure transparency in model predictions.
- Contribute to advancing sustainable waste management solutions through the integration of IoT and machine learning.

2. Methodology

In this study, two key approaches were employed that includes hardware implementation and machine learning experimentation. In hardware implementation, a dedicated sensor system has been developed to monitor composting emissions (e.g., methane, carbon dioxide, sulfur dioxide, and nitrogen oxide) and environmental conditions. This system, integrated with Arduino Mega 2560 R3 and ESP-32 microcontrollers, wirelessly transmits data to ThingSpeak for remote monitoring, ensuring safety through meticulous sensor calibration. In a machine learning experiment, our framework predicts composting maturity by analyzing features (temperature, C/N ratio, ammonia concentration, pH, nitrate content) from 10 datasets. After pre-processing and model training with five-fold cross-validation, we optimize hyperparameters using GridSearchCV. Transparency is prioritized with LIME and SHAP, making complex models interpretable. Here, both sections have been explained in detail.

2.1. Hardware implementation

2.1.1. Sensor integration of accurate monitoring

To independently measure methane, carbon dioxide, sulfur dioxide emissions, and nitrogen oxide, as well as temperature and humidity levels during composting processes, a dedicated sensor system has been developed to present a proactive response to these challenges. This system integrates various sensors, each specializing in detecting specific gases, temperature, or humidity, and establishes a connection to a pair of microcontrollers, namely the Arduino Mega 2560 R3 and ESP-32 (Fig. 1).

Through this interconnected setup, the collected data is transmitted wirelessly via Wi-Fi to the IoT platform known as ThingSpeak. Consequently, the results are visually represented in the form of graphs within ThingSpeak, enabling remote monitoring and eliminating the risk of exposure to toxic emissions. The following table outlines the sensors chosen for this purpose, along with their respective specifications:

The sensors are interconnected using soldered jumper cables on the PCB, which establish connections between the sensors and pins on the Arduino Mega 2560 R3 and ESP32 as shown in Fig. 2.

Within the Arduino Mega 2560 R3, the analog-to-digital conversion process handles readings from the CH4, NO2, and CO2 sensors. To optimize the performance and sensitivity, a load resistor of 22k Ohms is selected, following the manufacturer's recommendation. Additionally, both the SO2 sensor and the temperature/humidity sensor utilize I2C data communication to transmit their data to the Arduino Mega 2560 R3. Powering these sensors and microcontrollers is accomplished using a 5 V, 3A voltage regulator, which connects to a 12 V adapter via the power jack. This voltage regulator efficiently supplies power to the 5 V and GND pins of both the sensors and the microcontrollers.

Additionally, the Arduino Mega 2560 R3 interfaces with the ESP32, facilitating data transmission through UART / Serial

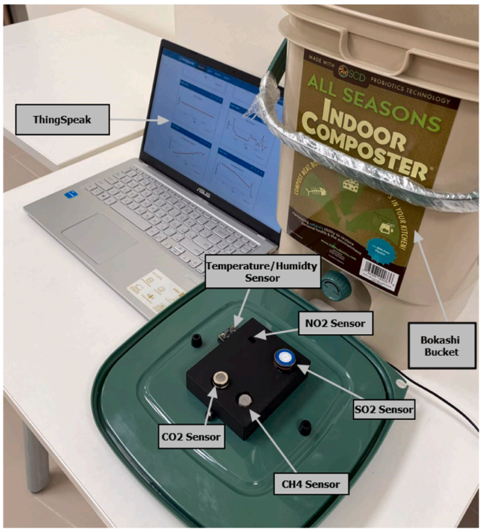


Fig. 1. Sensor system transmitting data to ThingSpeak. Sensor specifications have been added in the appendix (Table 1).

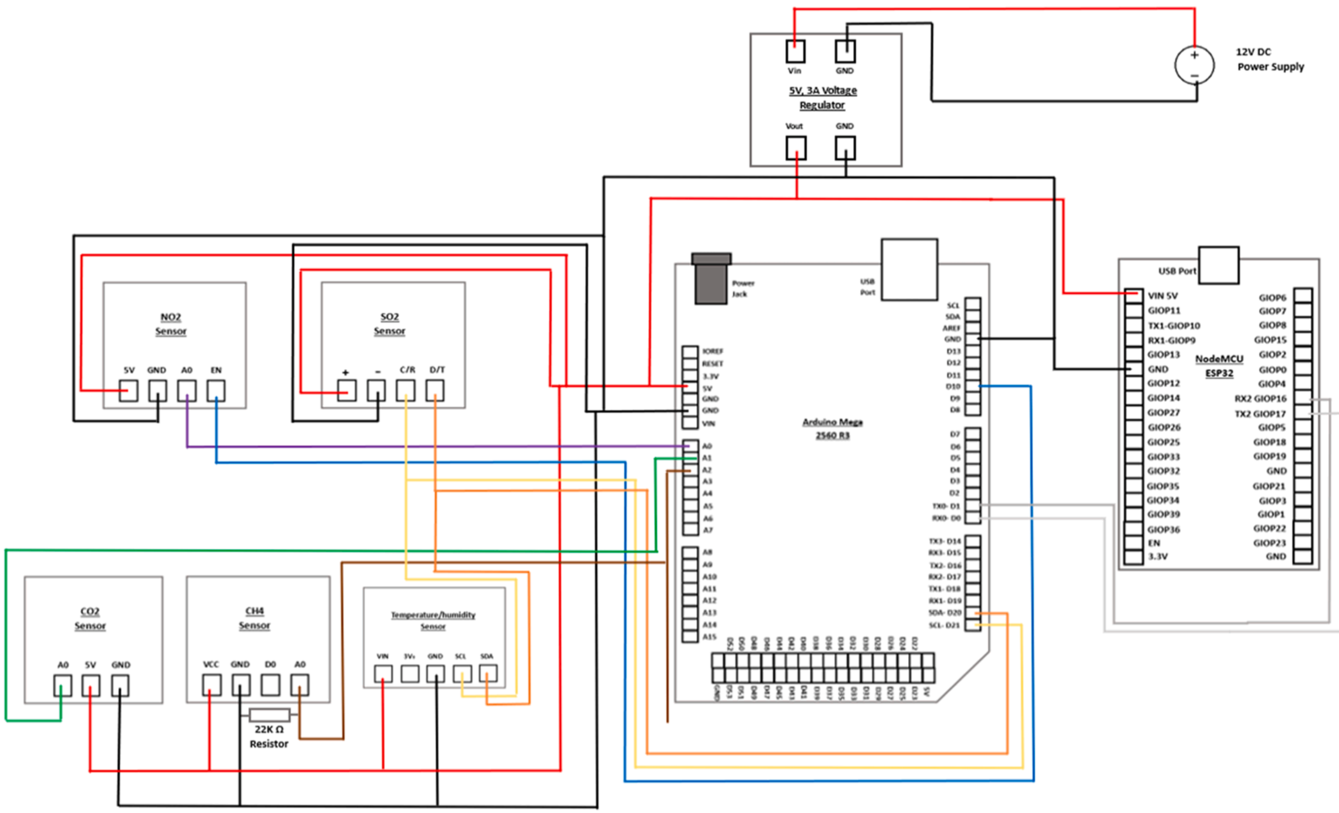


Fig. 2. Connection diagram of the sensor system.

communication. The choice of two microcontrollers stems from limitations within the ESP32, which directly measures analog voltages ranging between 0 V and 3.3 V, while the CO2 sensor outputs voltages from 2.7 V to 4.1 V, and the CH4 sensor ranges from 0.1 V to 4 .5V. Given these factors and the compatibility of sensor codes with the Arduino ecosystem, the Arduino Mega 2560 R3, equipped with multiple digital pins, including I2C, UART communication, and analog pins capable of reading in the range of 0–5 V, emerged as the ideal platform. Meanwhile, the ESP32 offers Wi-Fi data transmission capabilities to IoT platforms such as ThingSpeak. As a result, both microcontrollers establish a connected setup. These two microcontrollers interface with the computer through USB cables, linking to Arduino IDE software for code uploading. The table displaying the pin connections to link the sensors and ESP32 to the Arduino Mega 2560 R3 has been included in the appendix.

2.1.2. Sensor calibration for reliable data

To ensure the reliability of the sensor system, a meticulous calibration process is performed for various sensors, including SO_2, NO_2, CO_2, and CH_4. Essential factory calibration during production preserves the integrity of the sensor data, and it is worth noting that for the calibration of the SO_2 and NO_2 sensors, external libraries are employed: DFRobot_MultiGasSensor.h and DFRobot_MICS.h, respectively.

The calibration procedure for the CO_2 sensor involves a two-step approach. First, the sensor undergoes a 48-hour preheating phase to ensure consistent and stable operation. Subsequently, the sensor's output voltage was measured under ambient air conditions. Taking advantage of the ratio of the sensor's output voltage at 400 ppm and 1000 ppm CO_2 concentrations, as provided by the DFRobot website, the output voltage at 1000 ppm CO2 concentration is accurately determined. These calibration values are then integrated into the Arduino Mega 2560 R3. Similarly, the CH_4 sensor undergoes a meticulous calibration process. After a 48-hour preheating period, the MQ4 (CH_4) sensor calibration code from the provided appendix is uploaded to Arduino IDE software. The circuit arrangement, as illustrated in Fig. 2 (connection diagram), is adhered to, excluding the ESP32 connection. The MQ4 calibration code effectively computes the sensor's resistance in fresh air (R0), a crucial parameter in its performance. The calculated R0 value is then incorporated into the sensor system code for the Arduino Mega 2560 R3 sensor system code.

It is important to note that the NO_2 sensor requires 3 mins for calibration, during which no values are displayed on the serial monitor of the Arduino IDE software. Maintaining precision is paramount, and to achieve this, preheating the sensor system for 48 h before beginning data collection is essential. Once the preheating phase has concluded, the sensor system code for Arduino Mega 2560 R3 is uploaded to the Arduino IDE software. Upon successful upload, the ESP32 is interconnected with the Arduino Mega 2560 R3, as depicted in Fig. 2 (connection diagram). To enable data transmission, the ESP32 code is uploaded to a separate Arduino Mega IDE sketch, while at the sensor system, code for Arduino Mega 2560 R3 sensor system code continues to function. This configuration allows the ESP32 to effortlessly transmit data via Wi-Fi to a designated IoT platform, such as ThingSpeak (Fig. 3).

2.1.3. Implementing an IoT platform (ThingSpeak) for remote monitoring

To fulfill the need to remotely monitor gases, the ThingSpeak Internet of Things (IoT) platform was used for this project. This will protect individuals from the potential hazards posed by toxic emissions, which could otherwise lead to adverse health effects, such as respiratory complications. To transmit data to the designated IoT platform, ThingSpeak, the process involves creating a dedicated channel on the platform's website. Information regarding the channel creation process is included in the Appendix.

At the outset, the ESP32 establishes a connection with the local Wi-Fi network by inputting the appropriate network credentials into the provided ESP 32 code (see the Appendix). Subsequently, the ESP32 establishes communication using the platform's defined website as the host and the designated Write API key of the channel. This setup enables the ESP32 to upload gas measurements to the specified fields. The ESP32 efficiently stores the transmitted gas measurements from the Arduino Mega 2560 R3 in a buffer and then performs a conversion into double-precision floating values. These refined values are systematically transmitted to ThingSpeak's server at a defined interval of one second. Both the ESP32 code and the ThingSpeak platform indicate which specific field number corresponds to each gas. The values of the fields have been represented with a figure, which has been included in the appendix.

2.2. Machine learning experiment

After successfully setting up a reliable gas monitoring system and gathering data on gases (CH_4, CO_2, SO_2, NO_2), temperature, and humidity, we are moving on to the next step: using machine learning to make sense of these data. We will clean the data, create useful features, choose the right machine-learning models for compost maturity prediction, and evaluate their performance. This is how we will build our gas monitoring system.

2.2.1. Data collection

The data set used in this study was compiled from previous research efforts. It collected 452 samples with 11 attributes from various kitchen waste composting studies [14–16]. Of these, nine datasets were used to train and test various machine learning models. These datasets comprised a total of 416 samples, each characterized by 11 different attributes. These attributes covered a variety of factors, including composting time (day), temperature, moisture content (MC (%)), pH levels, carbon/nitrogen ratio (C/N ratio), levels of ammonia and nitrate (Ammonia (mg/kg) ', Nitrate (mg/kg) '), total nitrogen content (TN (%) '), total organic carbon content (TOC (%) '), electrical conductivity (EC (ms/cm) '),organic matter content (OM (%) '), T Value, and germination index (GI (%)). These attributes collectively contributed to the prediction of the concept of composting maturity.

In conjunction with the training dataset, a separate group of 36 samples was designated for external validation. This validation phase was aimed at validating the reliability and adaptability of the machine learning models in different settings.

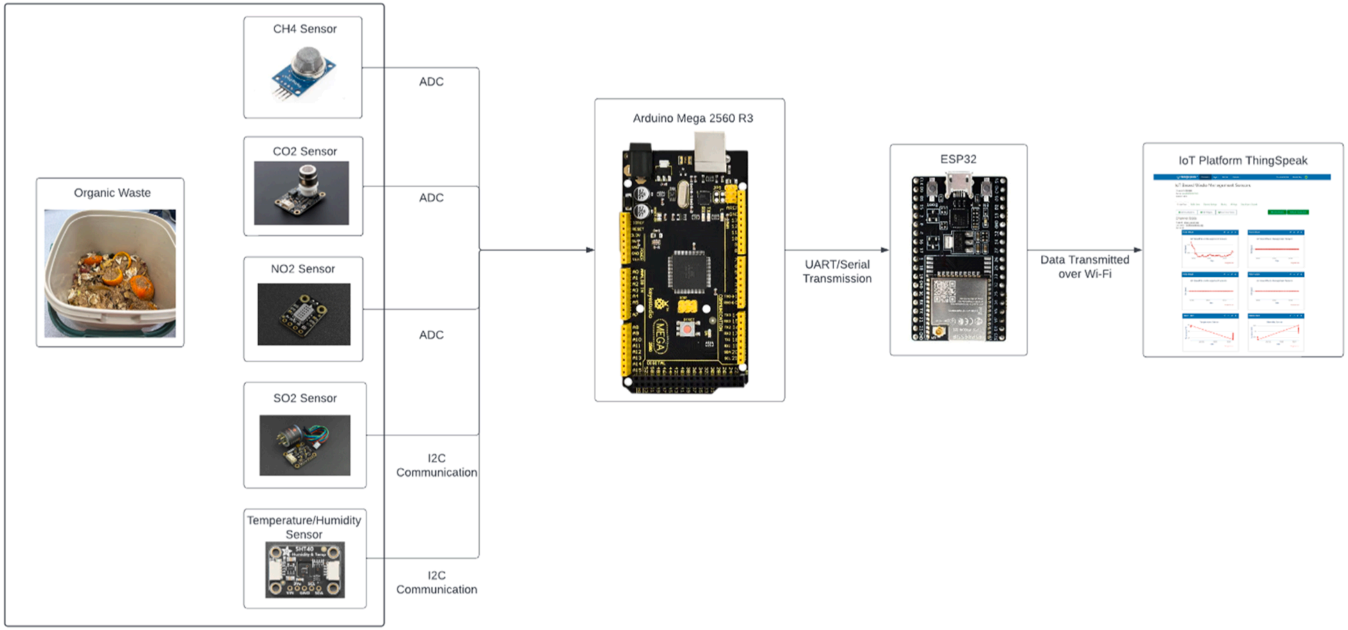


Fig. 3. Block diagram of the sensor system.

The core focus of this study is the understanding of composting maturity. This concept can be gauged through various indicators such as the level of humus, chemical properties, and potential harm to plants. When assessing the maturity of organic waste, two important aspects are considered. First, the stability of the composting material is examined. This is represented by changes in various substances such as cellulose, hemicellulose, carbon to nitrogen ratio (TC and TN), and the presence of humic acid (HA) [16]. In this study, the T value, which measures the carbon-to-nitrogen ratio over time, was chosen because of its relevance. It signifies the transformation of organic matter as composting progresses [15].

Secondly, the potential toxicity of the compost is taken into account. Early stages of decomposition release short-chain organic acids that can harm plants. With compost maturity, these harmful substances break down, reducing their phytotoxicity [13]. This is measured by the Germination Index (GI), which is a reliable indicator of compost maturity.

As the study progresses, the interaction between substance stability and toxicity emerges. This convergence is encapsulated in a combined score, reflecting both aspects. This comprehensive score encapsulates the essence of composting maturity, combining the aspects of substance stability and ecological balance.

2.2.2. Data preprocessing

Data preprocessing represents a cornerstone in the development of machine learning models, exerting a profound influence on predictive accuracy. The rigor and efficacy of preprocessing are directly proportional to the reliability of subsequent model predictions. In this study, the original data set was marked by the presence of numerous missing values, necessitating a robust interpolation strategy. Missing data was addressed by employing an interpolation method involving three nearest neighbors, a technique that enhances the reliability of the data set considering the inherent structure and relationships within the data.

After interpolation, standardization of the features was performed. The necessity for this step arises from the disparate range across features, which, if left uncorrected, can introduce bias, allowing certain variables to disproportionately influence model outcomes. Standardization, rather than normalization, was specifically chosen due to its robustness in handling outliers, a critical consideration given the nature of the dataset. This process ensures that each feature contributes proportionately to the final distance, thereby creating a balanced influence on the model's predictive function. The standardized values were computed according to Eq. (1).

$$y = \frac{x - \overline{x}}{\sigma} \tag{1}$$

Where, $\overline{x}$ = Mean of the data samples

σ = Standard deviation of the data samples

Through the intricate steps of this preprocessing pipeline, the data underwent a profound metamorphosis, aligned for intricate analysis and sowing the seeds of unbiased and optimistic predictions. The careful handling of missing values and the diligent standardization of feature magnitudes demonstrate the groundwork laid for accurate and reliable modeling. This pre-processing phase embodies the forefront of modern machine learning practices, solidifying its relevance in contemporary data science endeavors.

2.2.3. Proposed methodology

The framework presented in Fig. 4 of this study introduces an advanced approach to predict composting maturity. It analyzes various features such as temperature, C/N ratio, ammonia concentration, pH levels, and nitrate content, collected from different kitchen waste datasets. A total of 10 datasets are utilized: nine for model training and one for external validation. The initial steps involve handling missing data using the nearest neighbor method and standardizing variables for consistent scaling. A detailed feature analysis follows, shedding light on their significance and relationships. This sheds light on how waste quality influences composting attributes. The main phase involves training six machine learning (ML) models, employing a robust five-fold cross-validation to ensure broad applicability and prevent overfitting. Hyperparameters are fine-tuned using GridSearchCV for optimal model performance. The best model is then validated using data from an external dataset, demonstrating its reliability across diverse contexts. Transparency takes center stage, facilitated by LIME and SHAP. These tools elucidate how individual characteristics contribute to predictions, dispelling the opacity often associated with machine learning. By experimenting with feature parameters based on LIME and SHAP insights, the framework engages in an empirical exploration of feature impacts on composting maturity. This groundbreaking framework sets a new standard in predictive modeling. It blends cutting-edge techniques with a comprehensive approach, bridging scientific rigor and practical usability. The subsequent sections will dive into each step with precision and detail.

2.2.4. Data analysis

The analytical framework of this study is rooted in a meticulous and methodologically robust examination of the collection of characteristics. This pivotal process forms the foundation for achieving optimal model performance. Our exploration commenced with a dedicated evaluation of the significance of the features, harnessing the prowess of the XGBOOST model. This boosting framework, renowned for its precision and efficiency, illuminated the importance of each characteristic, as shown in Fig. 5. This revelation prompted a strategic choice to exclude a specific feature from further analysis. This strategic decision not only bolsters the computational efficiency of subsequent modeling but also has the potential to amplify the predictive accuracy of our models.

Deeper into our analytical exploration, we initiated an exhaustive correlation analysis, delving into the intricate interconnections among the diverse features and their links with the target variable of composting maturity. Employing the Pearson correlation

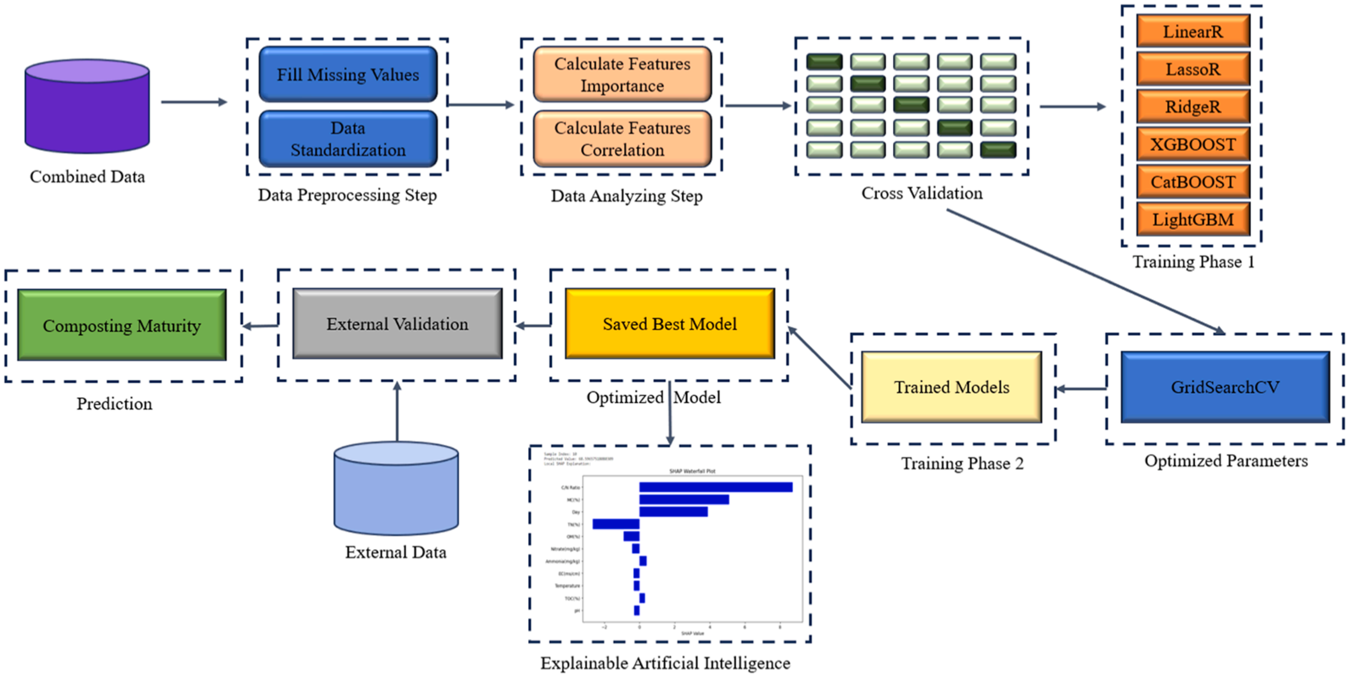


Fig. 4. Proposed framework for the prediction of composting maturity using optimized machine learning models with explainable artificial intelligence.

coefficient (PCC) as our chosen statistical metric, we methodically quantified these associations, as vividly depicted in Fig. 6. The histograms for each characteristic are demonstrated in Fig. 7.

2.2.5. Five-fold cross-validation and statistical measures

In this study, the adoption of a five-fold cross-validation (CV) methodology is a purposeful and scientifically sound choice for model training. This approach stands out for its ability to produce consistent and robust predictive insights that can be generalized effectively. This method, in contrast to the traditional train-test split, mitigates concerns about uneven data distribution or random allocation that can lead to inconsistent outcomes. The selection of a five-fold CV is a result of our keen awareness of the intricate nuances and potential biases that reside within the data. By segmenting the data set into five distinctive folds, we orchestrated a systematic rotation in which each fold took turns as the validation set, while the remaining folds constituted the training set. This meticulous cycling ensures that every data point participates in both the training and validation phases, thus leveling the playing field regarding data representation.

The essence of this approach hinges on its capability to rigorously scrutinize the robustness of the model. By subjecting the model to diverse training and testing scenarios across five iterations, we obtain a comprehensive understanding of its performance across various data subsets. This multipronged evaluation is pivotal in unveiling latent vulnerabilities and inconsistencies, fortifying our trust in the model's overall dependability and applicability. In our ongoing study, we used three distinct indicators, elucidated through Eqs. (2) to (4) [17].

$$\textit{Mean Absolute Error}\ (MAE) = \frac{1}{n}\sum_{i=1}^{n}\left|y_P - y_R\right| \tag{2}$$

$$\textit{Mean Squared Error}\ (MSE) = \frac{1}{n}\sum_{i=1}^{n}\left(y_P - y_R\right)^2 \tag{3}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_R - y_P\right)^2}{\sum_{i=1}^{n}\left(y_R - \overline{y}\right)^2} \tag{4}$$

Where, n = Total data samples, y_P = Predicted value, y_R = Actual value, and $\overline{y}$ = Mean of the actual values.

2.2.6. Developed machine learning models

In tackling the regression challenge at the core of this study, a curated ensemble of prominent state-of-the-art (SOTA) models has been assembled, each meticulously adapted to suit regression tasks. This assortment of models has been thoughtfully selected to encompass a spectrum ranging from conventional statistical methodologies to cutting-edge machine learning techniques. This curation aims to strike a harmonious balance between simplicity, interpretability, resilience, and predictive potency. Each model contributes distinct strengths to the analysis, facilitating the exploration of diverse facets of the underlying data relationships.

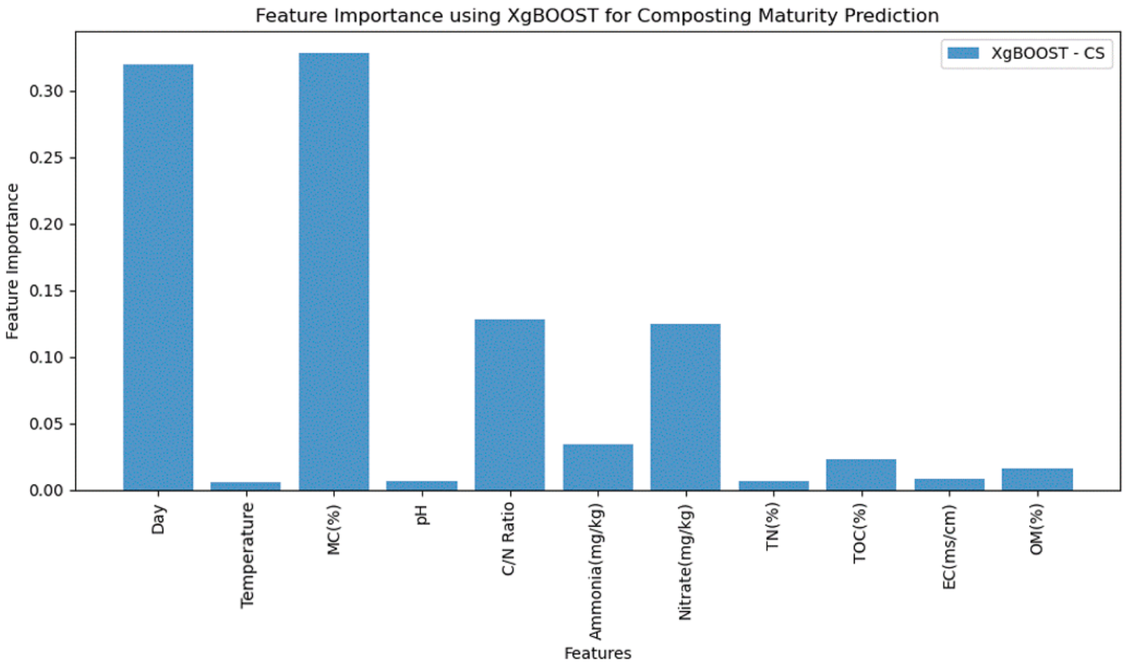


Fig. 5. Importance of features using XGBOOST to predict composting maturity.

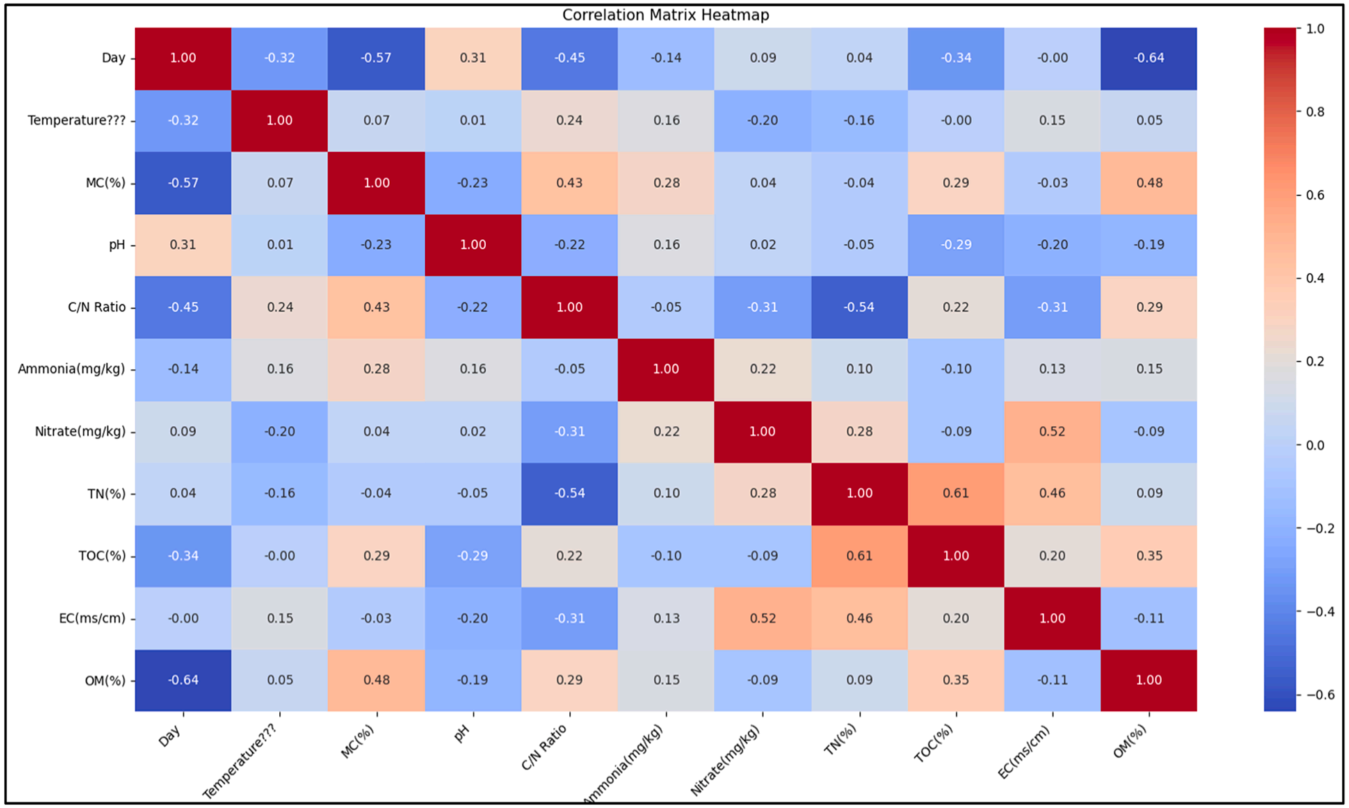


Fig. 6. Correlation between different characteristics.

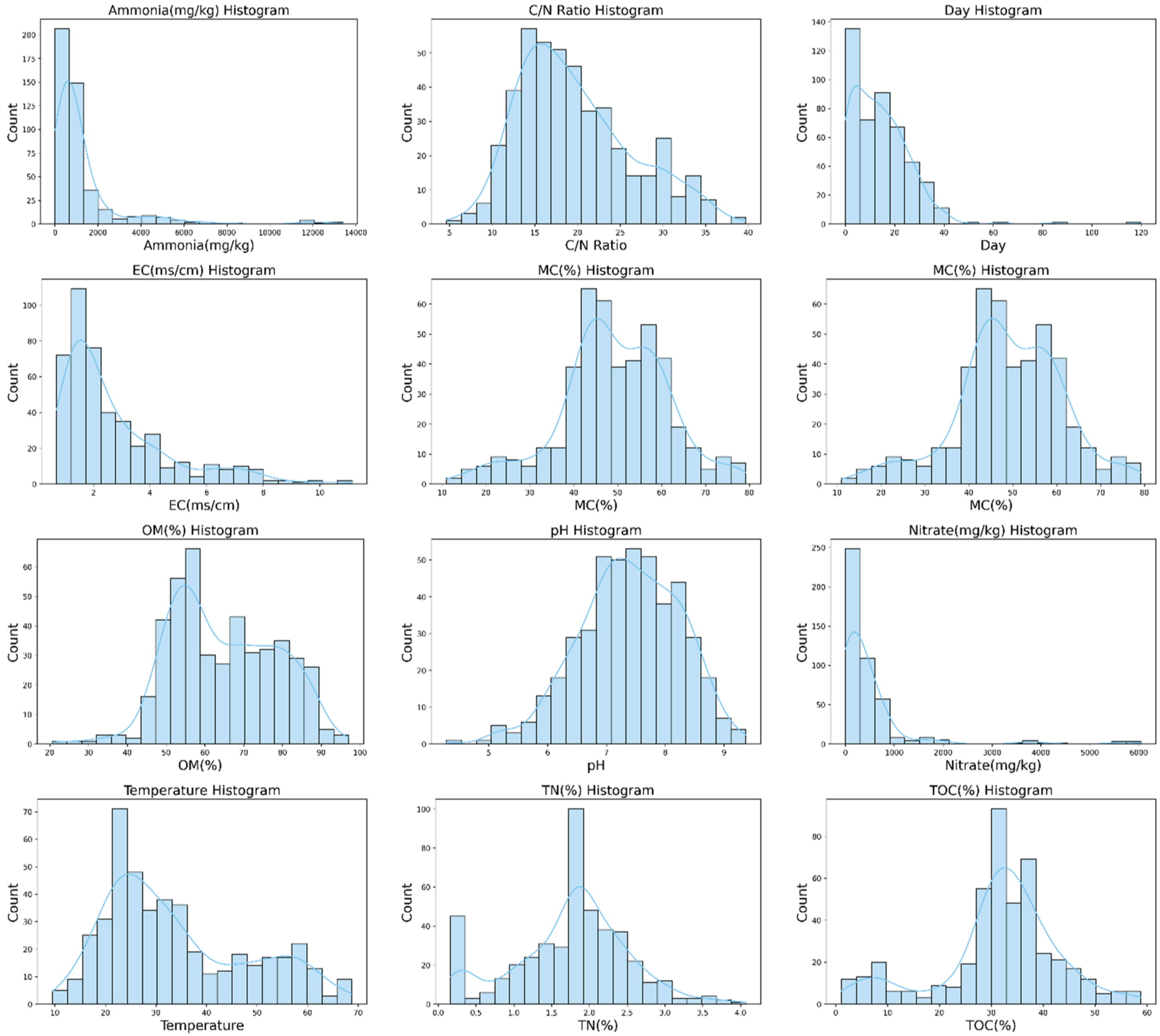


Fig. 7. Histograms for different features.

2.2.6.1. Linear regression. Linear regression serves as a fundamental tool for modeling the connection between independent and dependent variables through a linear equation [18]. The strength resides in its ease of interpretation and implementation. It assumes that a shift in a predictor yields a proportionate shift in the predicted response, rendering the relationships transparent. The capacity to extend this approach, such as employing polynomial regression, enables the modeling of nonlinear dynamics. Additionally, variants such as robust regression increase resilience against outliers and model inaccuracies. Its simplicity does come with limitations, as the assumption of linearity might not always hold. However, it stands as a pivotal baseline in diverse contexts.

2.2.6.2. Lasso regression. Lasso regression (Least Absolute Shrinkage and Selection Operator), a refinement of linear regression, introduces regularization by penalizing the absolute values of coefficients [18]. This regularization of L1 promotes sparsity, effectively removing certain coefficients. The strength of regularization is modulated by a hyperparameter, striking a balance between fitting and complexity. Lasso regression is particularly advantageous when dealing with superfluous or irrelevant features, offering automatic feature selection. This trait proves valuable in high-dimensional scenarios, where striking a balance between interpretability and generalization is paramount.

2.2.6.3. Ridge regression. Similarly to Lasso, Ridge regression incorporates regularization into linear regression, employing an L2 penalty on the coefficients [18]. This technique curtails the coefficients' magnitudes without enforcing them to become zero. The versatility of Ridge lies in its ability to span a continuum of models, ranging from standard linear regression to models featuring only the intercept. Ridge finds its niche in scenarios with correlated predictors, where conventional linear regression might falter. By adjusting the extent of the shrinkage through parameter tuning, Ridge Regression provides a robust estimation of model parameters, navigating the delicate equilibrium between model complexity and predictive efficacy.

2.2.6.4. XGBOOST. XgBoost, a paradigm in gradient boosting, orchestrates the sequential construction of multiple decision trees [19]. Each tree corrects the errors of its predecessor, culminating in refined predictions. Distinguished by its utilization of second-order gradient information, XgBoost enhances conventional gradient-boosting methodologies. Regularization terms embedded within its objective function prevent overfitting, and the model adeptly handles missing data. Enhanced attributes like early stopping, parallel processing, and innate support for sparse data underscore XgBoost's operational efficiency. Its triumph in numerous data science competitions underscores its real-world utility.

2.2.6.5. CatBOOST. CatBOOST, meticulously designed to grapple with categorical features, integrates ordered boosting [20]. By permuting the dataset, it mitigates the target leakage and protects against data contamination. Employing symmetric decision trees known as oblivious trees, CatBOOST applies identical splits to each instance, thereby fostering resilience to overfitting. This model boasts robust performance, demonstrating a diminished sensitivity to hyperparameter tuning. It simplifies pre-processing by autonomously handling categorical variables. Its visual interpretability tools, speed, and reliable performance make it an attractive choice across an array of applications.

2.2.6.6. LightGBM. LightGBM, another luminary in gradient boosting, is known for its prowess in both memory and computational efficiency [21]. The hallmark of LightGBM is its histogram-based algorithm, which discretizes continuous feature values into discrete bins. This innovation accelerates training without compromising accuracy. The decision to grow trees leaf-wise, rather than level-wise, increases precision for a given number of leaves. Augmented by GPU acceleration and adept handling of categorical features, LightGBM scales exceptionally well and shines as a dependable asset in large-scale data science endeavors.

2.2.7. Hyperparameters tuning using GridSearchCV

Within the realm of hyperparameter tuning through GridSearchCV, lies an element involving an exhaustive exploration of a predefined subset of an algorithm's hyperparameter space [22]. This process entails delineating a grid over the hyperparameters, ensuring the systematic assessment of every plausible combination within the defined range. Although this method leaves no possibility unexplored, its success hinges on the strategic selection of the parameter grid, maintaining the equilibrium between computational feasibility and optimization precision.

In the advancement of predictive modeling, meticulous calibration of hyperparameters emerges as a key factor influencing overall performance, interpretability, and model robustness. This study embraces a comprehensive hyperparameter tuning strategy via GridSearchCV, an approach that seamlessly amalgamates the virtues of methodical grid search with the robustness of cross-validation. This strategy was applied to several cutting-edge algorithms:

Using a five-fold cross-validation within the confines of GridSearchCV, each amalgamation within these grids was rigorously evaluated. This approach not only maximized the model's predictive accuracy but also ensured its performance evaluation on unseen data. With five-fold cross-validation on a dataset of 452 samples, each fold uses 80 % of the data (361 samples) as the training set and the remaining 20 % (91 samples) as the test set. This process is repeated across all five folds, ensuring every sample is included in both the training and test sets at different stages. This thorough and strategically orchestrated tuning process, tailored to the distinct characteristics of each model, aligns seamlessly with the study's commitment to analytical rigor, methodological innovation, and the pursuit of excellence in predictive modeling. It underscores the research's nuanced approach to model selection, reflecting an astute comprehension of the unique prospects and challenges posed by the data, thus enriching the frontiers of statistical learning and artificial intelligence.

The hyperparameter tuning for each model was conducted using GridSearchCV with five-fold cross-validation. This ensured a comprehensive evaluation of all parameter combinations to achieve optimal predictive performance. The specific settings for each model are summarized in the Tables 1 and 2:

2.2.8. Leveraging LIME and SHAP for Explainable AI

In the dynamic realm of intricate machine learning models, the introduction of LIME (Local Interpretable Model-agnostic Explanations) has ushered in a transformative era of Explainable AI (XAI), effectively bridging the gap between cutting-edge performance and comprehensible interpretability [23]. LIME's innovation lies in its adept ability to dissect complex models, unveiling the inner workings while preserving their predictive prowess. This is achieved by crafting a localized interpretation in proximity to a specific prediction, thus prioritizing individual predictions over the holistic behavior of the model. The LIME approach involves selecting a target instance for elucidation, perturbing it to curate a localized dataset, and subsequently training an interpretable model, such as LightGBM in our study, on this localized subset. Through this strategic methodology, LIME skillfully unveils the contributions of distinct features to specific predictions. This unveiling not only unravels the enigmatic rationales of black-box models but also lays bare potential biases or incongruities residing within them.

What sets LIME apart is its remarkable model-agnostic essence, which makes it adaptable to any flavor of machine learning model, from intricate deep neural networks to sophisticated ensemble methods. Operating as a unified conduit for translating cryptic model logic into transparent insights, LIME forms the cornerstone of a paradigm where the realms of explainability and performance seamlessly intertwine. In the landscape of our study, LIME becomes a symbol of empowerment, allowing practitioners and stakeholders to embrace AI with renewed trust, fortified integrity, and a greater ability to comprehend the inner workings of predictive systems.

Moreover, in tandem with LIME, the incorporation of SHAP (SHapley Additive exPlanations) augments our journey into the realm of explainable AI [24]. SHAP, similar to LIME, aspires to decode the intricate decision-making of complex models. It operates by attributing feature contributions to individual predictions, facilitating the dissection of a model's behavior. The use of SHAP in our study amplifies the transparency and interpretability of our AI systems, ensuring that predictive outcomes are not only accurate but also comprehensible. By synergistically embracing LIME and SHAP, we forge an unbreakable link between explainable AI and optimal performance, fostering a paradigm where the potential of AI is harnessed with unwavering confidence and insightful transparency.

3. Experimental results

3.1. Results of state-of-the-art models without GridSearchCV

This section focuses on a comprehensive comparative analysis of the outcomes obtained from six advanced machine learning models: linear, lasso, and ridge regression, along with XGBOOST, CatBOOST, and LightGBM. These models were harnessed in the context of composting maturity prediction. The data set encompassed 416 instances, distributed evenly in five folds, with each fold accommodating around 84 samples for testing and training.

Our experimentation yielded intriguing results. Concerning the prediction of composting maturity, the lasso regression (LR) exhibited comparatively lower performance, attaining the lowest R^2 value of 0.9045 in the fifth fold. Additionally, LR recorded the highest MSE and MAE values of 14.0975 and 1.8392, respectively, during the third fold, culminating in an average R^2, MSE, and MAE of 0.9149 ± 0.0079, 11.8926 ± 4.1137, and 4.34368 ± 0.43670, respectively. In contrast, the CatBOOST model showcased remarkable prowess by achieving an impressive R^2 of 0.9811 in the first fold. However, the LightGBM model emerged as the most consistent performer, securing the highest average R^2 of 0.97252 ± 0.0039. Meanwhile, the XGBOOST model demonstrated a minimum MSE of 2.5122 in the fifth fold and the lowest MAE of 1.2482, also in the same fold. Although the XGBOOST model had the lowest average MSE of 4.65252 ± 2.1713, the LightGBM model maintained the smallest average MAE of 1.68458 ± 0.1553.

It is evident from Table 3 that no single model emerged as the leader in both best R^2 performance and minimal errors. However, the LightGBM model showcased an optimistic R^2 and the least MAE, thus displaying accurate predictions of composting maturity.

Table 1
Specifications of gas sensors used in the Sensor System.

Sensor	CH4	CO2	SO2	NO2	Temperature and Humidity
Model	MQ 4	SEN0159 DFRobot	SEN0470 DFRobot	MEMS Gas Sensor MiCS-2714 SEN0441	SHT40 Adafruit
Operation Voltage DC	4.9–5.1 V approx. 150mA	3.7–5 V >500mA	3.3–5.5 V <5mA	4.9–5.1 < 20mA	3.3–5V
Measurement Range	200–10,000 ppm	0–10,000 ppm	0–20 ppm	0.05–10 ppm	0 °C-75 °C 0–100 % RH
Standard Operation temperature	18 °C-22 °C	−20 °C-50 °C	−20 °C-50 °C	−30 °C-85 °C	−40 °C-125 °C
Standard Operation Humidity	60–70 %	0–95 % RH (No condensation)	15–95 % RH (No condensation)	5–95 % RH (No condensation)	0–100 % RH

Table 2
Hyperparameters and their descriptions of the machine learning models.

Model	Hyperparameters	Hyperparameter Description
Linear Regression	fit_intercept: [True, False]	Controls whether the model includes an intercept term (bias).
Ridge Regression	alpha: [0.01, 0.1, 1.0]	Specifies the strength of L2 regularization to avoid overfitting.
Lasso Regression	alpha: [0.01, 0.1, 1.0]	Defines the L1 penalty strength, encouraging sparsity in coefficients.
XGBoost	n_estimators: [50, 100, 150], learning_rate: [0.01, 0.1, 0.2], max_depth: [3, 4, 5]`	Determines the number of boosting rounds, learning rate, and tree depth for gradient boosting.
CatBoost	n_estimators: [50, 100, 150], learning_rate: [0.01, 0.1, 0.2]`	Sets the number of boosting iterations, learning rate, optimizing for categorical features.
LightGBM	n_estimators: [50, 100, 150], learning_rate: [0.01, 0.1, 0.2], max_depth: [3, 4, 5]	Adjusts boosting iterations, learning rate, and tree depth, optimizing LightGBM's efficiency.

3.2. Results of state-of-the-art models with GridSearchCV

In this section, we discuss the influence of hyperparameter optimization on the performance of machine learning models used to predict fish length and weight. While our ML models demonstrated commendable results in terms of R^2, MAE, and MSE, we aimed to explore whether further enhancements could be achieved through a deliberate selection of model parameters rather than a random approach.

To achieve this goal, we harnessed the power of GridSearchCV to identify the most advantageous parameters for each ML model. The results of this optimization process are presented in Table 4. Interestingly, for XGBOOST, CatBOOST, and LightGBM, the optimal parameters remained consistent, determined through GridSearchCV.

Table 3
Fold-wise results for state-of-the-art models without GridSearchCV. (SD means standard deviation).

Model Name	Fold No	Composting Maturity		
		R^2 (↑)	MSE (↓)	MAE (↓)
Linear Regression	Fold 1	0.9590	6.0020	1.6006
	Fold 2	0.9532	6.9460	1.8272
	Fold 3	0.9530	9.0450	1.7622
	Fold 4	0.9440	14.0975	1.8392
	Fold 5	0.9600	5.5218	1.4279
	Average mean (μ) ± SD (σ)	0.95384 ± 0.0064	8.32246 ± 3.4998	1.69142 ± 0.1753
Lasso Regression	Fold 1	0.9120	9.1323	4.1261
	Fold 2	0.9212	10.5263	4.1311
	Fold 3	0.9123	11.1445	5.1221
	Fold 4	0.9245	19.1134	4.2124
	Fold 5	0.9045	9.5464	4.1267
	Average mean (μ)±SD (σ)	0.9149 ± 0.0079	11.8926 ± 4.1137	4.34368 ± 0.4367
Ridge Regression	Fold 1	0.9590	6.0020	1.6006
	Fold 2	0.9532	6.9460	1.8272
	Fold 3	0.9530	9.0450	1.7622
	Fold 4	0.9440	14.0975	1.8392
	Fold 5	0.9600	5.5218	1.4279
	Average mean (μ) ± SD (σ)	0.95384 ± 0.0064	8.32246 ± 3.4999	**1.69142** ± **0.1753**
XGBOOST	Fold 1	0.9685	3.2932	1.9231
	Fold 2	0.9623	5.1023	1.2482
	Fold 3	0.9726	4.2312	3.1247
	Fold 4	0.9632	8.1237	2.1980
	Fold 5	0.9712	2.5122	1.9497
	Average mean (μ) ± SD (σ)	0.96756 ± 0.0046	**4.65252** ± **2.1713**	2.08874 ± 0.6779
CatBOOST	Fold 1	0.9811	3.3423	2.0144
	Fold 2	0.9623	6.3457	2.3045
	Fold 3	0.9411	15.8974	2.8134
	Fold 4	0.9395	10.9485	2.9043
	Fold 5	0.9563	5.8231	1.9485
	Average mean (μ) ± SD (σ)	0.95606 ± 0.0171	8.4714 ± 4.9779	2.39702 ± 0.4435
LightGBM	Fold 1	0.9754	9.4116	1.7122
	Fold 2	0.9723	4.6958	1.9133
	Fold 3	0.9765	7.3015	1.4866
	Fold 4	0.9665	8.6123	1.6907
	Fold 5	0.9719	6.4620	1.6201
	Average mean (μ) ± SD (σ)	**0.97252** ± **0.0039**	7.29664 ± 1.8483	1.68458 ± 0.1553

*Bold values indicate the best results.

In the realm of composting maturity prediction, the lasso regression (LR) exhibited a relatively lower performance, echoing our earlier findings, with the lowest R^2 value of 0.9336 was observed in the fourth fold, as demonstrated in Table 5. Furthermore, LR recorded the highest MSE and MAE values of 17.1923 and 2.9922, respectively, during the third fold, contributing to an average R^2, MSE, and MAE of 0.9366 ± 0.0059, 10.1636 ± 4.0966, and 2.81814 ± 0.2672, respectively. On the contrary, the CatBOOST model showed impressive prowess, achieving a notable R^2 of 0.9937 in the first fold. Meanwhile, the XGBOOST model demonstrated consistent performance, securing the highest average $R^2$of 0.98702 ± 0.0043. Furthermore, the CatBOOST model demonstrated the lowest MSE of 1.8382 in the first fold and the smallest MAE of 1.1845 in the fifth fold. Furthermore, the XGBOOST model exhibited the lowest mean MSE and MAE of 4.08896 ± 1.7951 and 1.3714 ± 0.1584, respectively.

In conclusion, a meticulous process of hyperparameter optimization has showcased the superior predictive capacity of the XGBOOST model in composting maturity prediction. These findings strengthen the reputation of the XGBOOST model as a highly promising tool for such tasks, setting a remarkable benchmark for future research in this field. Its exceptional predictive accuracy and robustness, as evident from our results, establish it as a formidable model for research in waste management science and beyond.

3.3. Performance comparisons between models with and without optimized parameters

This section emphasizes the comparative analysis of diverse machine learning models, both in their original configurations and after optimization using GridSearchCV. The central objective was to investigate the performance improvements attributed to GridSearchCV implementation in the prediction of composting maturity.

In the context of the prediction of compost maturity, the results, as depicted in Table 6, reveal that the performance of the linear and ridge regression models experienced minimal alteration after optimization. However, notable progress was observed in the LR model, with R^2 surging from 0.9245 to 0.9435, indicating an approximate 2 % increase. Consequently, a distinct decrease was observed in MSE from 19.1134 to 10.0352 (approximately 9 % decrease in error) and in MAE from 4.2124 to 2.8923 (approximately 2 % decrease in error). These results signify a substantial reduction in errors after parameter optimization, resulting in improved model performance. This underscores the significant merits of employing the GridSearchCV technique to refine machine learning models.

Furthermore, the performance of the XGBOOST model also witnessed enhancement, with R^2 escalating from 0.9726 to 0.9912, translating to an almost 2 % increase, while both MSE and MAE exhibited reductions.

In summary, this comparative assessment elucidates the profound influence of GridSearchCV optimization on the efficacy of machine learning models. However, certain limitations were observed, particularly with models like Lasso Regression, which showed marginal improvement in performance despite optimization, likely due to the non-normal distribution of features in the dataset. Additionally, linear regression models, while improved, still exhibited higher errors compared to ensemble methods like XGBoost and CatBoost, indicating their constraints in handling complex, non-linear relationships. These insights underscore the importance of selecting appropriate models and recognizing their limitations to ensure balanced and effective solutions. Our findings particularly highlight the potential of optimized models to significantly improve accuracy in predicting composting maturity, offering valuable tools for waste management departments to streamline processes and reduce operational costs while emphasizing the need for continued research to address existing constraints.

3.4. Friedman statistical test analysis

The Friedman test has been done for a comprehensive evaluation of the models' performance across multiple aspects. R^2, MAE, and MSE were selected to provide a balanced assessment. R^2 measures the model's goodness of fit, MAE assesses the average error, and MSE highlights large errors. These metrics together offer a complete view of the models' performance, covering fit, error magnitude, and sensitivity to larger errors. The ranks for each algorithm across MSE, MAE and are as follows:

The Friedman test is applied to statistically evaluate whether significant differences exist among the algorithms. Based on the rankings across all metrics in the Table 7 (R^2, MSE, and MAE), the rank orders for the algorithms vary, reflecting differences in their performance across these metrics. This variation underscores the importance of using multiple metrics to provide a comprehensive comparison of the models.

Friedman Test Calculation

The formula for the Friedman test statistic is as follows:

Table 4
Best parameters selected by GridSearchCV.

Model/Parameters	Learning Rate	No of Estimators	Fit Intercept	Alpha	Max Depth
Liner Regression	–	–	True	–	–
Lasso Regression	–	–	–	0.1	–
Ridge Regression	–	–	–	1.0	–
XGBOOST	0.2	150	–	–	5
CatBOOST	0.2	150	–	–	–
LightGBM	0.2	150	–	–	5

Table 5
Fold-wise results for state-of-the-art models with GridSearchCV. (SD means standard deviation).

Model Name	Fold No	Composting Maturity		
		R^2 (↑)	MSE (↓)	MAE (↓)
Linear Regression	**Fold 1**	0.9590	6.0020	1.6006
	Fold 2	0.9532	6.9460	1.8272
	Fold 3	0.9530	9.0450	1.7622
	Fold 4	0.9440	14.0975	1.8392
	Fold 5	0.9600	5.5218	1.4279
	Average mean (μ) ± SD (σ)	0.95384 ± 0.0064	8.32246 ± 3.4998	1.69142 ± 0.1753
Lasso Regression	**Fold 1**	0.9330	7.1220	2.3446
	Fold 2	0.9425	8.9461	2.9373
	Fold 3	0.9435	10.0352	2.8923
	Fold 4	0.9336	17.1923	2.9922
	Fold 5	0.9304	7.5222	2.9243
	Average mean (μ) ± SD (σ)	0.9366 ± 0.0059	10.1636 ± 4.0966	2.81814 ± 0.2672
Ridge Regression	**Fold 1**	0.9590	6.0020	1.6006
	Fold 2	0.9532	6.9460	1.8272
	Fold 3	0.9530	9.0450	1.7622
	Fold 4	0.9440	14.0975	1.8392
	Fold 5	0.9600	5.5218	1.4279
	Average mean (μ) ± SD (σ)	0.95384 ± 0.0064	8.32246 ± 3.4998	1.69142 ± 0.1753
XGBOOST	**Fold 1**	0.9890	3.1909	1.2834
	Fold 2	0.9835	4.3065	1.3172
	Fold 3	0.9900	3.4857	1.5178
	Fold 4	0.9814	7.0597	1.5541
	Fold 5	0.9912	2.4020	1.1845
	Average mean (μ) ± SD (σ)	0.98702 ± 0.0043	**4.08896 ± 1.7951**	**1.3714 ± 0.1584**
CatBOOST	**Fold 1**	0.9937	1.8382	2.8909
	Fold 2	0.9847	4.0061	1.0406
	Fold 3	0.9620	13.2717	1.7292
	Fold 4	0.9788	8.0299	1.1884
	Fold 5	0.9923	2.1054	0.9177
	Average mean (μ) ± SD (σ)	**0.9823 ± 0.01285**	5.85026 ± 4.8309	1.5336 ± 0.8093
LightGBM	**Fold 1**	0.9799	5.8392	1.5622
	Fold 2	0.9740	6.7950	1.7882
	Fold 3	0.9747	8.8432	1.7147
	Fold 4	0.9643	13.5328	1.7960
	Fold 5	0.9805	5.3481	1.3852
	Average mean (μ) ± SD (σ)	0.97468 ± 0.0065	8.07167 ± 3.3336	1.64926 ± 0.1749

*Bold values indicate the best results.

Table 6
Performance comparisons with and without GridSearchCV for composting maturity prediction.

Model	Without GridSearchCV			With GridSearchCV		
	R^2(↑)	MSE (↓)	MAE(↓)	R^2(↑)	MSE (↓)	MAE(↓)
Linear Regression	0.9600	5.5218	1.4279	0.9600	5.5218	1.4279
Lasso Regression	0.9245	19.1134	4.2124	0.9435	10.0352	2.8923
Ridge Regression	0.9600	5.5218	1.4279	0.9600	5.5218	1.4279
XGBOOST	0.9726	4.2312	3.1247	0.9912	2.4020	1.1845
CatBOOST	0.9811	3.3423	2.0144	0.9937	1.8382	2.8909
LightGBM	0.9765	7.3015	1.4866	0.9805	5.3481	1.3852

*Bold values indicate the best results.

Table 7
Friedman Ranking of Models Based on Performance Metrics.

Model	R^2 Rank	MSE Rank	MAE Rank	Sum of Ranks
Linear Regression	4.5	3.5	3.5	11.5
Lasso Regression	6	6	6	18
Ridge Regression	4.5	3.5	3.5	11.5
XGBoost	1	1	1	3
CatBoost	2	2	5	9
LightGBM	3	4	2	9

$$X_F^2 = \frac{12N}{K(K+1)}\left(\sum_{i=1}^{k} R_i^2\right) - 3N(k+1)$$

Where:

- N represents the number of datasets, (since ranks are averaged, assume N is the number of metrics, which is 3: MSE, MAE, R^2).
- k denotes the number of algorithms, set to 6.
- R_i indicates the total ranks assigned to the i th algorithm.

Sum of all squared ranks: 758.5

After converting the values into the Friedman test statistic formula:

$$\text{Calculated Friedman Statistic } \left(X_F^2\right) = 588.35$$

Critical Value:

The critical value for the Friedman test, at a significance level of α=0.05 with 5 degrees of freedom ($k-1$, where $k=6$), is approximately 11.07, based on the chi-squared distribution.

Finding:

Since the calculated Friedman test statistic ($X_F^2 = 588.35$) is much greater than the critical value (11.07), we reject the null hypothesis.

This means there is a statistically significant difference in the ranks of the models based on their performance across R^2, MSE, and MAE. Some models significantly outperform others. Rejecting the null hypothesis indicates a statistically significant difference in the ranks of the models based on their performance across R^2, MSE, and MAE. This suggests that some models significantly outperform others. The advantage of rejecting the null hypothesis is that it allows for the identification of superior models, which can guide future optimization and model selection. It also ensures that the observed differences are not due to random chance, thereby strengthening the validity of the model comparison.

Based on the Friedman test and rank sums, the top three models are XGBoost, CatBoost, and LightGBM. XGBoost has the lowest rank sum of 3, indicating it outperformed the other models across R^2, MSE, and MAE. CatBoost and LightGBM both have rank sums of 9, making them the next best models. These results suggest that XGBoost is the best-performing model, followed closely by CatBoost and LightGBM.

3.5. *Unraveling the generalization capability of optimized ML models: external validation*

This segment presents strong evidence of the remarkable generalizability of the machine learning models optimized through our GridSearchCV method. Originally trained on a combined dataset, we retained the best-performing models for validation in an external context. This external dataset, consisting of 36 samples from different environments, was used to assess the robustness and predictive prowess of these models. As illustrated in Table 8, all models exhibited outstanding performance in forecasting composting maturity. In particular, the XGBOOST model once again emerged as the frontrunner. It achieved an impressive R^2 score of 0.9892, accompanied by low values of MSE and MAE of 1.9127 and 1.0659, respectively.

These findings firmly establish the effectiveness of our proposed machine learning models, affirming their capability to deliver precise forecasts when applied to new and practical real-world situations. With their high precision in predicting composting maturity, these models have the potential to significantly improve waste management. This could reduce the dependency on manual labor, facilitate large-scale operations, and maintain cost-effectiveness. In essence, this study provides substantial support for the applicability of our innovative machine-learning techniques in various practical scenarios, offering an economical solution to enhance the global feasibility and sustainability of waste management practices.

3.6. *Performance comparisons with SOTA models*

In this particular section, we conducted a comprehensive comparison between our proposed best-performing model and the most

Table 8
Performance of optimized ML models on an external data set.

Model	Composting Maturity		
	R^2 (↑)	MSE (↓)	MAE (↓)
Linear Regression	0.9754	4.9931	1.8434
Lasso Regression	0.9823	3.1484	1.3655
Ridge Regression	0.9711	4.8761	1.9001
XGBOOST	**0.9892**	**1.9127**	**1.0659**
CatBOOST	0.9870	4.0890	1.3714
LightGBM	0.9817	3.2479	1.4554

*Bold values indicate the best results.

recent State-of-the-Art (SOTA) model available. During our investigation, we discovered only one study that had ventured into predicting kitchen composting maturity using various machine-learning models. This study used the same dataset, a fusion of 10 different data sets, for its experimentation. Their approach involved allocating 70 % of the data for training and the remaining 30 % to test their models. Our focus was primarily on assessing how our models compared to theirs.

In this specific part of our research, we conducted a thorough comparison between our innovative model and the most advanced existing model in the field, often referred to as the State-of-the-Art (SOTA) model. We found that only one study had previously attempted to predict kitchen composting maturity using different machine-learning techniques. This study used the same data set as ours, which combined data from ten different sources. They divided the data for training and testing, using 70 % for training and 30 % for testing. Our focus was on evaluating how our model performed compared to theirs.

After analyzing their results in Table 9, we noticed that their most successful model was a stacking model, achieving an R^2 value of 0.8746. They also reported a mean squared error (MSE) of 6.4539 and a mean absolute error (MAE) of 3.7936. Our exploration with the same dataset showcased a notably higher R^2value of 0.9892, marking an approximately 11-fold improvement over their model's performance. Furthermore, through careful optimization, our XGBOOST model outperformed theirs by considerably reducing the MSE to 1.9127, a reduction of nearly five times. Similarly, our XGBOOST model achieved a significantly lower MAE of 1.0659, which was almost four times smaller than its recorded value.

However, it is important to note that the study by Ding et al. did not produce results for T value and GI, crucial factors in evaluating compost maturity [15]. This gap was addressed by the study by Yalin et al., where their fusion model achieved R^2values of 0.928 for the GI and 0.957 for T value [25]. However, their model did not predict overall composting maturity. The work of Wan et al.'s work displayed the best $R^2$of 0.977 for GI among all SOTA models [26].

Our approach used the well-optimized XGBOOST model exclusively to predict the T value and GI. Impressively, our proposed model achieved a high $R^2$of 0.985 for GI, surpassing Wan et al.'s result by almost 1 %, and an $R^2$of 0.979 for T value, which was about 2 % better than Yalin et al.'s model. These results demonstrate that our proposed XGBOOST model is more robust compared to other current top-performing models, especially in terms of the T value, the GI, and the prediction of composting maturity. A key insight emerges from the unique ability of our model, which effectively overcomes the limitations present in existing SOTA models. In particular, these established models have a significant drawback. They struggle to predict complete values that encompass the GI, the T value, and the broader spectrum of composting maturity. This limitation has resulted in a fragmented landscape, where certain models solely predict GI, while others exclusively focus on forecasting the composting maturity score. Another study [27] utilized a comprehensive methodology to analyze critical parameters influencing green waste compost maturity through machine learning models, specifically employing Extra Trees, Gradient Boosting, Multi-Layer Perceptron (MLP), and KNN on a dataset of 638 samples. Key outputs indicated that the Extra Trees model effectively predicted the Germination Index (GI) and T value, with time being the most critical parameter affecting compost maturity, while humic acid and the C/N ratio also played significant roles. However, the study acknowledged limitations such as reliance on potentially biased datasets, the complexity of model interpretation, and the need for further research to enhance generalizability across diverse composting scenarios.

In light of these limitations, our research introduces an innovative approach, a groundbreaking method that transcends these challenges by providing accurate predictions for both the GI and the T value. These factors, which play a crucial role in assessing composting maturity, form the core of our contribution. Through our inventive methodology, we seamlessly integrate the prediction of these vital metrics, infusing new vitality into the field, and propelling us into unexplored frontiers of prediction accuracy.

These outcomes have significant implications. They underscore the robustness and effectiveness of our fine-tuned XGBOOST model in making accurate predictions about composting maturity. Our model outperforms the existing SOTA model by substantial margins across various evaluation metrics. This comparison underscores the credibility and dependability of our proposed approach, showcasing its potential to offer advanced solutions to predict composting outcomes with greater precision.

Our proposed XGBOOST model exhibits higher computational complexity due to iterative boosting and GridSearchCV but achieves superior accuracy compared to simpler models like Ding et al.'s stacking model. This ensures precise predictions, making it ideal for applications requiring high accuracy despite increased computational demands.

3.7. Explainable artificial intelligence

Explainable Artificial Intelligence (XAI) represents a significant advancement within the realm of machine learning, offering an unprecedented opportunity to unravel the complexities inherent in intricate models. This study marks a pioneering effort by employing LIME and SHAP techniques to unravel the fundamental elements that impact composting maturity [28].

Table 9
Performance comparisons with SOTA models.

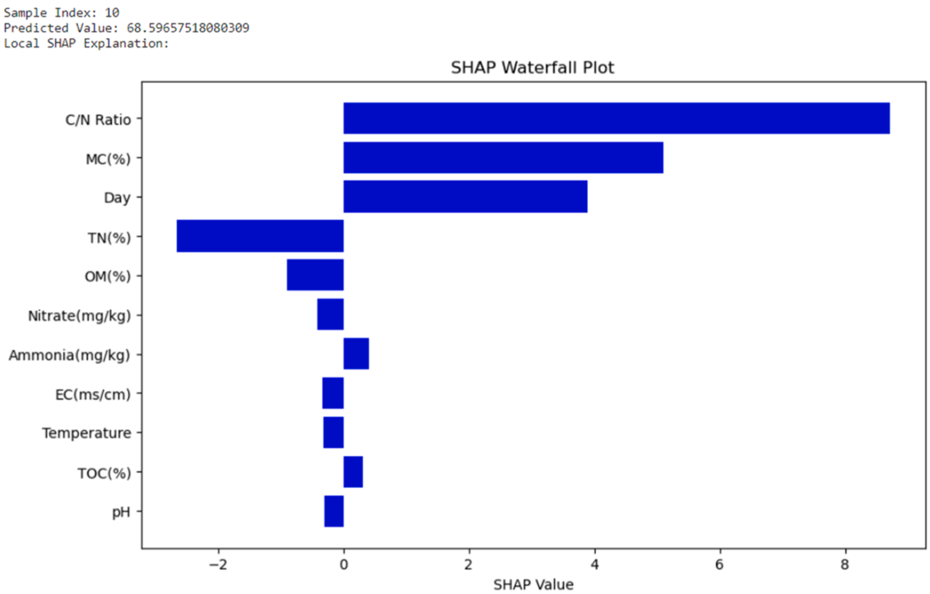
Authors	Composting Maturity Prediction			GI Prediction			T Value Prediction		
	R^2(↑)	MSE(↓)	MAE(↓)	R^2(↑)	MSE(↓)	MAE(↓)	R^2(↑)	MSE(↓)	MAE(↓)
Ding et al. [2]	0.8746	6.4539	3.7936	–	–	–	–	–	–
Yalin et al. [12]	–	–	–	0.928	–	–	0.957	–	–
Wan et al. [13]	–	–	–	0.977	–	–	–	–	–
Our work	**0.9892**	**1.9127**	**1.0659**	**0.985**	**2.341**	**1.098**	**0.979**	**4.788**	**2.164**

*Bold values indicate the best results.

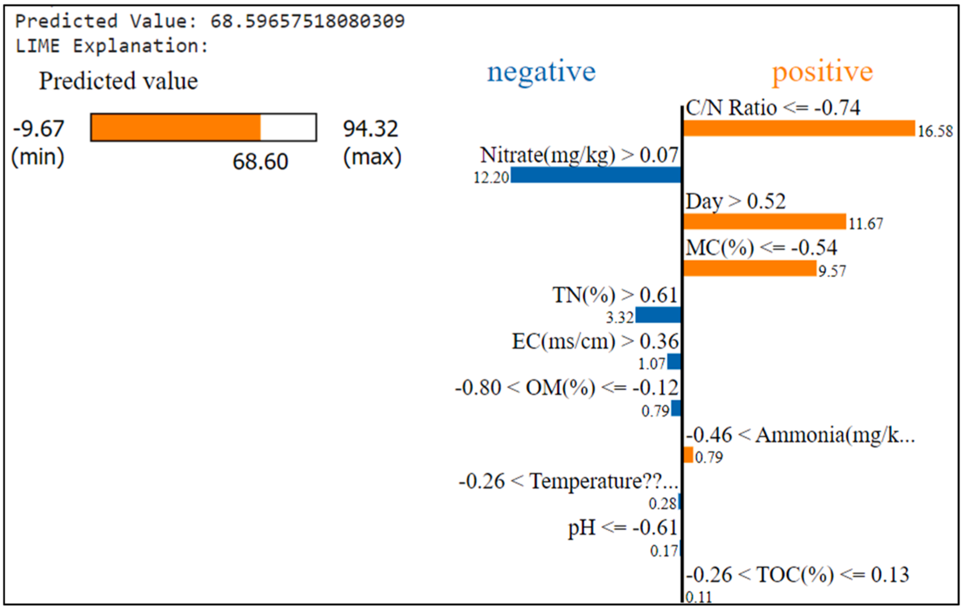
In many cases, machine learning models are regarded as enigmatic 'black boxes,' where the specific contributions of individual features to predictions remain obscured [28]. To tackle this challenge, our study has innovatively integrated LIME and SHAP into the XGBOOST framework. This unique combination aims to identify the key characteristics that drive improvements in the maturity of kitchen compost, enabling people without deep expertise to comprehend the intricacies of machine learning in aquaculture applications.

In Fig. 8, we visually portray the intricate connections among different features that contribute to predicting composting maturity. This graphical representation serves as a straightforward way to understand the factors that influence the journey toward composting maturity. Among these features, we find some notable players: time, the balance of the C/N ratio, nitrate levels, moisture content (MC), ammonia levels, and the presence of total organic carbon (TOC). These elements work together to guide the process of composting towards its eventual state of maturity.

When we dig deeper into our findings, we notice that the passage of time and the equilibrium of the C/N ratio emerge as key



(A)



(B)

Fig. 8. Explainable AI for the prediction of composting maturity using (A) SHAP and (B) LIME.

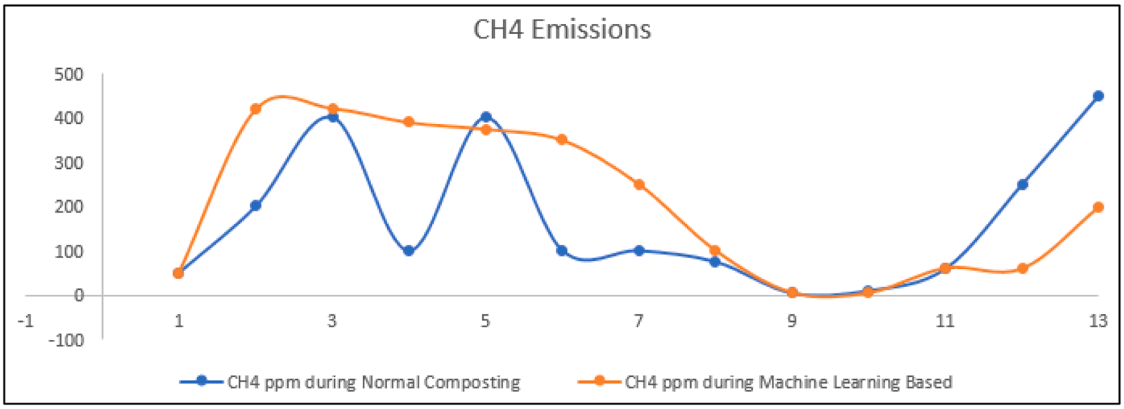
influencers, significantly steering composting towards higher levels of maturity. These insights align well with the contributions we see in Fig. 5, which are effectively highlighted by the predictive capabilities of the XGBOOST algorithm. On the flip side, attributes such as total nitrogen (TN), organic matter (OM), and pH exert a contrasting influence, tending to slow the progression toward composting maturity.

In support of these findings, the SHAP analysis underscores the consistency of these characteristic dynamics. This graphical representation proves to be a useful tool, offering an accessible interface that does not require specialized expertise. The interplay of these attributes revealed through XAI, captures the essence of expediting composting processes. The triumvirate of day, MC, and C/N ratio rise as pivotal factors, steering the course toward maturity, a reflection of their crucial roles in shaping the journey.

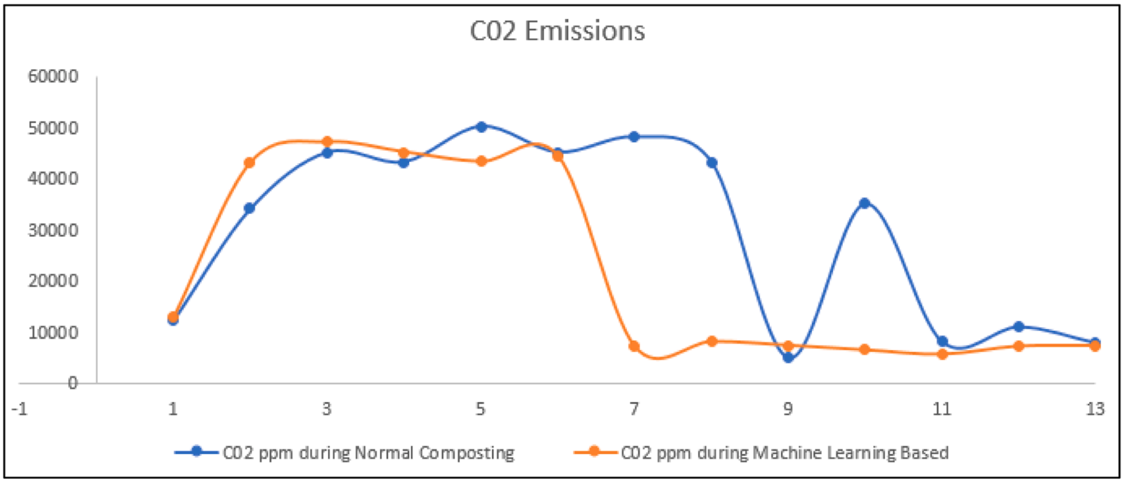
Drawing on these insightful revelations, our study underscores the potential of XAI, especially when combined with the transparent and versatile XGBOOST framework. This partnership goes beyond just enhancing predictive abilities, as seen in improved R^2 values and lower loss metrics (MSE and MAE). The real brilliance lies in the clarity this fusion provides: Equipping operators with a practical tool. By grasping the nuances of individual parameters, operators can navigate the maturity of composting, thus influencing profitability. Significantly, this empowerment does not require specialized knowledge or labor-intensive management efforts.

These advances push XAI into a sphere of transformative impact, ushering in operational finesse and greater profitability. This shift, rooted in the realm of waste management, sends ripples far beyond, reshaping possibilities and opening new avenues for innovation.

Fig. 9 illustrates the gas emissions of CH_4 and CO_2 during the composting process, comparing the normal maturity of the composting with predictions made using machine learning. In particular, the figures show that the differences between normal composting and machine learning-based predictions are not very large, and at certain points, they align completely. This indicates that machine learning can accurately predict gas emissions. The value of machine learning predictions lies in their ability to provide early insight into gas emissions. By closely monitoring and adjusting key parameters, such as temperature, moisture, and aeration in real time, the system can encourage aerobic decomposition, ultimately reducing methane generation.



(A)



(B)

Fig. 9. Gas emission graph for normal composting maturity prediction and machine learning-based prediction using (A) CH_4 and (B) CO_2.

In our study, both methane (CH_4) and carbon dioxide (CO_2) emissions peaked around the seventh day of composting due to our optimized approach, which speeds up compost maturation. Machine learning helped predict and manage these emissions. Our experiments also confirmed that there were no harmful SOX and NOX emissions, and the use of machine learning reduced gas emissions, contributing to more sustainable composting processes. Furthermore, it is important to note that our experiment did not produce harmful SOX and NOX emissions, as verified by the sensors integrated into the composting process. This reaffirms the environmentally friendly nature of the composting operation.

This approach offers several notable advantages, including continuous real-time tracking of emissions, which allows early detection and corrective actions [26]. Furthermore, machine learning can forecast future emissions, optimize composting conditions, and potentially reduce environmental impact. Additionally, it can lead to cost savings, support data-driven decision-making, and ensure compliance with regulations, making it a valuable tool for enhancing the efficiency, sustainability, and eco-friendliness of composting operations.

3.8. Real-world validation of enhanced composting: synergizing machine learning and explainable artificial intelligence

In a tangible embodiment of our research, we translated theoretical insights into practical action as Ding et al. [15], solidifying the profound synergy between our ML model and the enlightening revelations of XAI. This empirical journey materialized through the careful mixing of kitchen waste and wood chips in a harmonious mass ratio of 8:2, forming the basis of our composting materials. Two distinct cohorts were established: the Control group (CK), which underwent conventional direct mixing, and the Modified group (ML), subjected to the methodology meticulously outlined in our XAI section.

The inception of this practical voyage involved a meticulous preheating regimen utilizing a controlled water bath, prepping the materials for subsequent adjustments. The intricate calibration proceeded with finesse: The moisture content was judiciously set at around 65 %, and the pH values were carefully aligned to a steady 6.82, achieved by introducing 0.6 % lime into the mix. Furthermore, total nitrogen (TN) levels were purposefully adjusted to approximately 2.2 %, a feat accomplished by precise incorporation of soybean meal in calculated proportions. The meticulous orchestration of variables revealed that additional optimization beyond day 14 was not requisite, as both TN and total organic carbon (TOC) elegantly converged within optimal thresholds. The experimental conditions remained constant throughout the study.

The empirical result emerged with distinction: the CK group produced a residual mass of 6.12 kg after composting, while the ML group displayed a significantly reduced mass of 4.98 kg. This translates into proportional reduction rates of 38.9 % and 50.1 %, respectively, a testament to the heightened degradation experienced by waste within the ML group.

In terms of practical findings, the cornerstone of our study found fruition in increasing the carbon-to-nitrogen (C/N) ratio and Moisture Content (MC) through strategic interventions. This was achieved by the calculated incorporation of various elements, which synergistically influenced these parameters, catalyzing the composting process. This real-world validation stands as a testament to the efficacy of our sophisticated ML model, fortified by the insights extracted from XAI. The harmony between empirical outcomes and model predictions not only accentuates the efficiency of composting practices but also propels the realm of waste reduction towards a sustainable horizon.

3.9. Structure of bacterial communities in composting environments

The journey through the intricacies of composting unfolds as we dive into the captivating world of bacterial communities at the genus level (Fig. 10). Within this microcosm, Firmicutes emerge as the undisputed rulers, their dominance extending their firm grip on both the mesophilic and thermophilic stages of the composting process [15]. In the spotlight of the mesophilic stage, Lactobacillus and Leuconostoc take center stage, orchestrating a captivating performance. These remarkable genera reveal their ability to thrive during the storage of kitchen waste, magically producing organic acids. The foundation of our insight lies in a meticulous tabular representation of microbial counts, providing a portal to the intricacies of microbial life. Distinct microbial species and genera are illuminated in the first row of this tableau, encompassing the likes of Lactobacillus, Bacillus, Pseudogracilibacillus, and a multitude of others.

Two distinct sample types, cryptically labeled CK and ML, serve as our canvases, likely signifying diverse experimental setups or methodologies. Our voyage across time was delineated through five distinct temporal waypoints: day 0, day 4, day 10, day 20, and day 30. These temporal markers hint at the multifaceted stages of the experiment, where the microbial saga unfolds. Within this tableau, numerical enigmas represent microbiological counts, portraying the abundance of each microbial species or genus within the corresponding samples at specific temporal junctures.

As we embarked on our analysis, we uncovered a multitude of intriguing observations:

In the CK samples, the charismatic Lactobacillus takes the stage with an initial flourish, boasting a maximum concentration of 70. However, as time unfurls its relentless tapestry, this genus's concentration gracefully wanes, eventually reaching the vanishing point by day 30. Gleichzeitig, the counts of other microbial protagonists, such as Bacillus, Lactococcus, and Clostridium, pirouette through time, each executing its unique dance of change. The microbial tapestry of the ML samples unveils a distinct narrative, weaving a different story from the CK counterparts. On day 0, Proteus, Acinetobacter, and Lysinibacillus grace the stage, their populations embarking on a mesmerizing journey of ebb and flow throughout the temporal expanse. A remarkable absence on day 0, Pseudomonas, emerges dramatically, scaling to a crescendo of 50 by day 30.

The composition of the microbial community morphs dynamically over time, as some species ascend to greater prominence, others fade into obscurity, and a select few remain resolute in their presence. Zero counts at certain temporal waypoints and for specific

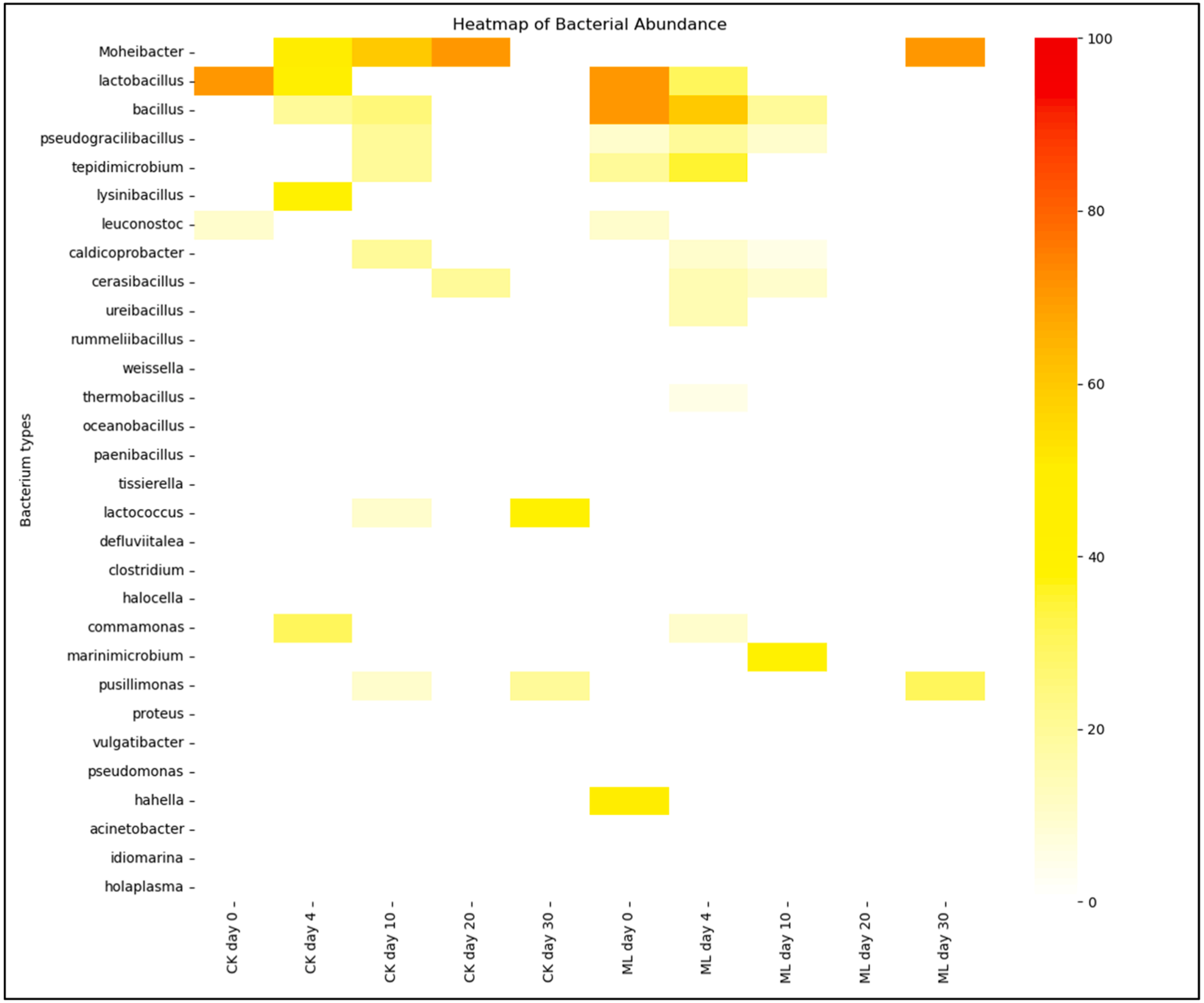


Fig. 10. Proportional presence of the most prominent bacterial groups at the genus level throughout the composting process.

species poignantly signify their absence in the corresponding samples during those moments.

In the grand tapestry of composting, the ML group shines with greater microbial diversity within an optimized environment. Furthermore, the microbial ensembles that drive organic matter transformation exhibit a predilection for early development, exemplified by Bacillus, and abundant accumulation, embodied by Marinimicrobium and Vulgatibacter, among others. The stage is set for an enchanting exploration of the structure of bacterial communities as they navigate the labyrinthine pathways of the microbial landscape.

4. Conclusion

Effective monitoring of gas emissions during composting processes is crucial for sustainable waste management and minimizing environmental impacts. Both aerobic and anaerobic composting provide eco-friendly options for the treatment of organic waste but release gases in the process. Our dedicated sensor system and IoT integration enable accurate real-time data collection and remote analysis, ensuring responsible waste management that protects the environment and human well-being. As we improve our understanding of gas emissions, we pave the way for a more sustainable future. Through machine learning, we predicted compost maturity using the collected data. Our analysis before and after GridSearchCV optimization emphasizes the importance of parameter tuning, it significantly improved the results for all algorithms tested, leading to enhanced model performance. In particular, XGBOOST achieved the highest R^2 value of 0.9912, while CatBOOST exhibited the lowest mean squared error (MSE) at 1.8382. Furthermore, XGBOOST demonstrated the lowest mean absolute error (MAE) of 1.1845. The improved performance, as demonstrated by XGBOOST and other algorithms, underscores the importance of optimized models for enhancing waste management efficiency and reducing costs, propelling future advances in waste management systems. Understanding the maturity of compost enables us to take appropriate actions to improve it, ensuring optimal compost quality for agricultural and environmental applications. Future work will focus on developing advanced sensors to capture real-time environmental data and testing their accuracy across diverse composting conditions. Real-time monitoring systems will be explored for continuous tracking of compost maturity and gas emissions. Multisite validation will ensure the scalability and robustness of the approach, aiming to create an efficient system for sustainable waste management.

While the proposed sensor-based machine learning approach demonstrated strong predictive performance, several limitations must be acknowledged. Sensor reliability can be affected by environmental factors and sensor drift, which may lead to inaccuracies in gas emission data and compost maturity predictions. Additionally, the models, though effective in most cases, faced challenges with extreme environmental conditions or unbalanced datasets, where predictions were less reliable. The technique's generalizability to different composting materials beyond the specific kitchen waste and wood chips mixture used in this study may also require further calibration. Additionally, the computational complexity of these models could pose limitations for real-time application in large-scale composting operations without further hardware optimization. These constraints suggest that future improvements could focus on refining sensor accuracy, optimizing computational efficiency, and ensuring the approach's broader applicability to diverse composting scenarios.

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Author contributions

Conceptualization: Amith Khandakar, Azad Ashraf; Methodology: Amith Khandakar, Mohamed Arselene Ayari, Azad Ashraf, Amin Esmaeil Jahanand Aljarrah, Philips Michael; Investigation: Md. Nahiduzzam, Hafsa Binte Kibria, Vasiliki Maria Gerokosta, Abdul Ahad Shehbaz, Maryam Abdulla R A Al-Mansoori, Farah Khattab; Writing – Original Draft: Amith Khandakar, Md. Nahiduzzam; Writing – Review & Editing: All authors contributed to reviewing, editing, and formal analysis of the results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.compeleceng.2025.110115.

Data availability

The processed dataset used in this study can be accessed via the following link: https://github.com/hafsa-kibria/Compost-Dataset. For any further details or inquiries, please contact the corresponding author.

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